STRATIFICATION OF A FINITELY PRESENTED MORPHISM BY SMOOTHNESS LOCI

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 16011-16052, immediately following the statement at 15986-16009.

Editorial note. Two cross-chapter pointers (Limits) plus one same-chapter lemma are recorded in assembly.external_references.

Editorial note. The statement's descending chain notation with its conditions is retained verbatim in the statement span.

Editorial note. Steps the author leaves to the reader are flagged here for the reviewer: source line(s) 16024 carry an "omitted"/"follows easily"/"immediate from the definitions" phrase. Those sentences are retained verbatim; the displayed argument around them stays inside the counted span.

Allowed standard prerequisites (editorial).

- Smoothness stratification at a generic point. chapter lemma lemma-smoothness-stratification-at-generic-point, cited in the proof. Supplies the local step: after shrinking, the morphism becomes smooth over a nonempty open of the source.
- Approximation of a finitely presented morphism by Noetherian data. Limits, proposition limits-proposition-approximate and lemma limits-lemma-descend-finite-presentation, cited in the proof. Lets the Noetherian fppf-local statements be applied after approximating the situation.
- Noetherian induction on the source. standard argument used to remove each successive smoothness locus. Iterates the local step to produce the descending chain of closed subschemes.

Source reference. Original lemma 0H43: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 15886–15886. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01ZM: focused-seeds/geometry/S9-morph-smoothness-stratification/ref-tags.txt, source lines 4–4. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01ZA: focused-seeds/geometry/S9-morph-smoothness-stratification/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let $f : X \to S$ be a morphism of finite presentation between quasi-compact and quasi-separated schemes. Then there exists a $t \geq 0$ and closed subschemes*

$$S \supset S_0 \supset S_1 \supset \ldots \supset S_t = \emptyset$$

such that

(1) *$S_i \to S$ is defined by a finite type ideal sheaf,*
(2) *$S_0 \subset S$ is a thickening,*
(3) *for each i there exists a surjective finite locally free morphism $T_i \to S_i \setminus S_{i+1}$,*
(4) *for each i there exists a $t_i \geq 0$ and closed subschemes*

$$X_i = X \times_S T_i \supset Z_{i,0} \supset Z_{i,1} \supset \ldots \supset Z_{i,t_i} = \emptyset$$

such that $Z_{i,j} \to X_i$ is defined by a finite type ideal sheaf, $Z_{i,0} \subset X_i$ is a thickening, and such that the morphism $Z_{i,j} \setminus Z_{i,j+1} \to T_i$ is smooth.

Proof. We can find a cartesian diagram

$$\begin{array}{ccc} X & \longrightarrow & X_0 \\ \downarrow & & \downarrow \\ S & \longrightarrow & S_0 \end{array}$$

such that X_0 and S_0 are of finite type over $\mathbf{Z}$. See Limits, Proposition 01ZA and Lemma 01ZM. Thus we may assume X and S are of finite type over $\mathbf{Z}$. Namely, a solution of the problem posed by the lemma for $X_0 \to S_0$ will base change to a solution over S; details omitted.

Assume X and S are of finite type over $\mathbf{Z}$. In this case every quasi-coherent ideal is of finite type, hence we do not have to check the condition that S_i is cut out by a finite type ideal. Set $S_0 = S_{red}$ equal to the reduction of S. Let $\eta \in S_0$ be a generic point of an irreducible component. By Lemma 0H43 we can find an open subscheme $U \subset S_0$, a surjective, universally injective, finite locally free morphism $V \to U$, a $t_0 \geq 0$ and closed subschemes

$$X \times_S V \supset Z_{0,0} \supset Z_{0,1} \supset \ldots \supset Z_{0,t_0} = \emptyset$$

such that $Z_{0,i} \to X \times_S V$ is defined by a finite type ideal sheaf, $Z_{0,0} \subset X \times_S V$ is a thickening, and such that the morphism $Z_{0,i} \setminus Z_{0,i+1} \to V$ is smooth. Then we let $S_1 \subset S_0$ be the reduced induced subscheme structure on $S_0 \setminus U$. By Noetherian induction on the underlying topological space of S, we may assume that the lemma holds for $X \times_S S_1 \to S_1$. This produces $t \geq 1$ and

$$S_1 = S_1 \supset S_2 \supset \ldots \supset S_t = \emptyset$$

and t_i and $Z_{i,j}$ as in the statement of the lemma. This proves the lemma. □